Experiment 4. Random Walk Simulation

Aim: To simulate a random walk process

Procedure:

Step 1: Create Required MATLAB Files

- run.m
- fun.m
- add.m
- area_m.m
- randomwalkBoundary.m (main function)

Step 2: Set Working Directory

1. Open MATLAB
2. Navigate to the folder:

Step 3: Understand the Main Function

The main function call (from Command Window) is:

randomwalkBoundary(50,3,true);

Parameters:

- 50 → Number of steps (N)
- 3 → Number of dimensions (D)
- true → Enables plotting/animation

Step 4: Run the Simulation

In the **Command Window**, type:

close all
randomwalkBoundary(50,3,true);

Step 5: Execution Flow

1. The program initializes the walk
2. For each dimension:
   - It generates random steps
   - Updates position
3. A loop (for i=1:length(P)) progressively plots:
   - Step-by-step movement
   - Animated trajectory
4. pause(0.2) creates animation effect
5. Each dimension is plotted with different lines

Step 6: Output Visualization

- X-axis → Step number
- Y-axis → Position
- Graph shows **random movement over time**
- Multiple lines → Multiple dimensions

Step 7: Modify Inputs (Optional)

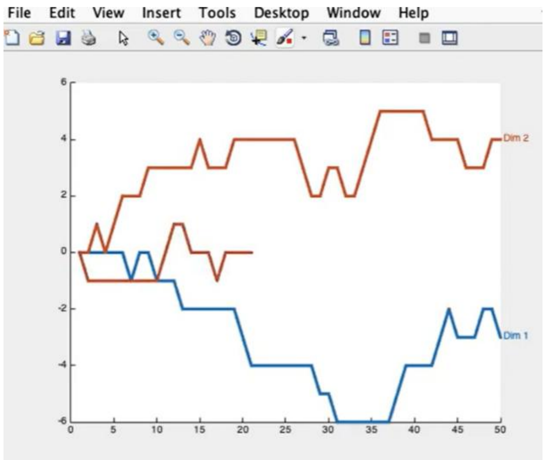
Try different values:

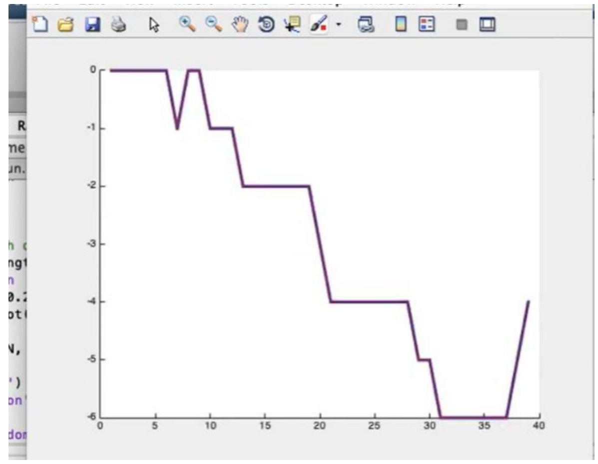
randomwalkBoundary(100,1,true); % 1D walk
randomwalkBoundary(200,2,true); % 2D walk
randomwalkBoundary(50,3,false); % Without animation

Code:

```
function P = randomwalkBoundary(N, D,
animate)

% RANDOM WALK WITH OPTIONAL
ANIMATION

% N → Number of steps

% D → Number of dimensions

% animate → true/false for animation


% Initialize position matrix

P = zeros(N, D);


% Generate random steps (-1 or +1)

steps = randi([0,1], N, D)*2 - 1;


% Compute positions

for k = 1:D

    for i = 2:N

        P(i,k) = P(i-1,k) + steps(i,k);

    end

end


% Plotting

figure;

hold on;


for k = 1:D

    for i = 1:N

        if animate

            pause(0.05);

        end

        plot(1:i, P(1:i,k), 'LineWidth', 2);

        hold on;

    end


    % Label each dimension

    text(N, P(N,k), sprintf(' Dim %d', k));

end


xlabel('Step #');

ylabel('Position');


if D == 1

    title('Random Walk in 1 Dimension');

elseif D == 2

    title('Random Walk in 2 Dimensions');

else

    title(['Random Walk in ', num2str(D), '
Dimensions']);

end


grid on;

end
```

Results:





Conclusion:

The MATLAB random walk simulation successfully demonstrates how a system evolves under **random step movements** across one or more dimensions. The animated plotting helps visualize the stochastic behavior clearly. As the number of steps increases, the path becomes more irregular and unpredictable, reflecting the nature of probabilistic processes. This type of simulation is widely useful in fields like physics (Brownian motion), finance (stock price modeling), and engineering systems analysis.